Citizen humanities: from theory to practice

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Introduction

Citizen science has become a transformative force in cultural heritage research, fostering collaboration between researchers and the public to tackle challenges in preservation, accessibility, and sustainability. Within the broader "citizen humanities" framework, this participatory approach bridges traditional humanities research with digital innovation and societal engagement, aligning with principles of openness and inclusivity (Longley Arthur & Hearn, 2021). This study investigates the role of citizen science in cultural heritage through an ontology-based analysis of projects. By examining their aims, tools, and outcomes in academic literature and databases, it highlights how participatory practices integrate technology and community input, democratizing data collection, promoting engagement, and driving interdisciplinary collaboration for sustainable heritage management.

Methodology

A systematic search in the Scopus database identified 44 relevant articles on citizen science projects in cultural heritage. Keywords "cultural heritage" and "citizen science" were combined using the Boolean operator (AND), with filters applied for English language and specific document types, such as research articles and conference papers. To enrich the findings, project websites were consulted for further information, and metadata, including timelines and funding, were extracted from the EU Cordis database. Projects were annotated using the Citizen Science Ontology and organized into an Excel sheet with ontology-based categories.

Aims

The aims of the projects demonstrate the expansive potential of citizen science in cultural heritage and environmental preservation. These initiatives prioritize accessibility through digital tools and open platforms, ensuring broader access to cultural resources. Community participation is central, involving efforts to conserve biodiversity, protect natural landscapes, and address threats to cultural heritage, including degradation and conflict. Education and active engagement empower communities to meaningfully contribute to preservation. Specialized areas, such as marine heritage, urban sustainability through smart technologies, and safeguarding intangible cultural elements like oral histories, also feature prominently. By integrating interdisciplinary methods and innovative tools, these projects foster collaboration across societal and institutional boundaries, uniting heritage preservation with environmental and community priorities.

Table 1: Aims of the projects

Aim	*Description*	Project Examples
Digital Heritage and Data Accessibility	Development of open-access digital services and tools for cultural heritage.	*DigiNUMA* , *Fragmentarium Project* , *FindSampo* , *SuALT* , *SigNet* .
Protection of Natural and Cultural Environment	Conservation of biodiversity, natural landscapes, and cultural heritage.	*Christmas Bird Count* , *A Red List of Terrestrial Isopods* , *Eurasian Kurgan Database* , *Pamvotis Lake Bioblitz* , *Earth Loving Explorers* , *Nautilos Project* , *SeaRise Citizen Science* .
Detection and Preservation of Heritage	Addressing the degradation of cultural heritage and implementing preventive measures for preservation.	*STORM Project* , *MIPAC-CM* , *COMPACT* , *Heritage Monitoring Scouts* , *Protection of Heritage in Conflict Zones* , *Documentation of Cultural Heritage in Raqqa (Syria)* .
Community Engagement and Education	Empowering and educating communities about heritage preservation through active participation.	*CITiZAN* , *HMS Florida* , *INNOVA Cuba* .
Marine and Underwater Heritage	Conservation and monitoring of marine ecosystems and underwater cultural heritage.	*Green Bubbles* , *GIRT Scientific Divers* , *Odyssea Blue Heritage Project* .
Smart Cities	Creating sustainable urban environments through the use of geographic information and citizen participation.	*Volunteer Geographic Information System (VGI)* .
Intangible Cultural Heritage and Memory	Safeguarding oral histories, local traditions, and intangible heritage through community input.	*Characterization of Meanings of Toponyms of Kenya* , *Ghost Signs Project* , *Wood Products for Culture and Heritage* , *Letters 1916* , *Dictionary of Borys Grinchenko* .

Tools

The tools used in citizen science projects for cultural heritage demonstrate a seamless blend of technology and community participation. These tools streamline data collection, management, and sharing, while advanced platforms enhance collaboration and efficiency. Technologies like GIS mapping, machine learning, and digital photogrammetry boost accuracy in heritage documentation and analysis. Participatory platforms and crowdsourcing actively involve communities, fostering inclusive preservation efforts. By integrating diverse tools, these projects tackle conservation challenges, promote education, and ensure the sustainable management of cultural and natural heritage.

Table 2: Tools for Cultural Heritage Projects

Category	*Tools*
Data Integration and Management	Sampo Model (RDF, SPARQL Endpoint, Linked Data Finland Platform), DaSCH Service Platform (DSP), LAMP Stack (Linux, Apache, MySQL, PHP), Omeka S, collaborative open-access databases, APIs for interoperable data sharing
Digital Photogrammetry and Visualization	3DF Zephyr Aerial, MeshLab, CloudCompare, GIS-based mapping tools, IIIF-compatible platforms for image annotations, ArcGIS StoryMaps, Django-based interactive maps
Geographic and Spatial Analysis	QGIS, ArcMap/ArcGIS, Python scripting, Leaflet API, HTML & JavaScript, OpenStreetMap integration, smartphones (GPS-equipped), real-time kinematic (RTK) tools
Machine Learning and AI	PIA Platform (object detection and co-embedding tools), AI-based annotation tools, search filtering methods
Crowdsourcing and Public Engagement	Zooniverse, PyBossa, SeaRise Beta, Anecdata, Google Forms, Audiboo, Zoom, SPOD Virtual Platform, online discussion groups
Educational Tools and Platforms	Online citizen training modules, NAS (Nautical Archaeology Society) Guides, UNESCO training manuals, mobile-friendly web apps, interactive maps
Cataloging and Archival Analysis	IIIF Standards (JSON-LD, IIIF Presentation API), Microsoft Document Applications, historical manuscripts, participatory annotation platforms (e.g., PIA), personal interviews with volunteers
Conservation and Preservation	CitPres (DataCollector and Map tools), OpenBioMaps, Django Framework, IIIF standards for interoperability, repositories for kurgans and other heritage assets
Audio-Visual Processing	Audiboo (citizen audio contributions), GIS-based tagging for soundscapes, dynamic visualization of multimedia content

Outcomes of the projects

The analyzed projects showcase the transformative impact of citizen science in cultural and natural heritage. Notable achievements include the discovery of medieval coins and Roman-era artifacts through metal detection, as well as the documentation of coastal heritage sites under threat. These initiatives address environmental and human-induced risks, such as graffiti and urban decay, by fostering community-driven conservation. Biodiversity mapping projects, such as those tracking pollinators and isopods, emphasize ecological contributions and inform preservation strategies. Additionally, linguistic and archival efforts have expanded digital collections, enhancing access to cultural assets and promoting public engagement. These measurable results underscore the effectiveness of citizen participation in tackling complex heritage challenges while ensuring long-term preservation and societal involvement.

Discussion

The analysis underscores the significant role of citizen science in advancing cultural heritage preservation and research. By involving the public, these initiatives democratize heritage practices and encourage inclusivity in addressing cultural and environmental challenges. Notable achievements include biodiversity mapping, urban sustainability efforts, and the creation of accessible digital archives. The integration of technologies like machine learning and GIS is instrumental in achieving these outcomes, while participatory platforms empower communities to actively contribute. Despite challenges such as sustainable funding and policy development, the findings highlight the potential of citizen-driven initiatives to shape resilient and collaborative approaches to cultural heritage preservation.

Bibliography

Longley Arthur, P., & Hearn, L. (2021). Toward open research: A narrative review of the challenges and opportunities for open humanities. *Journal of Communication*, *71*(5), 827–853.